

Lifting a corner involution to an ambient CAR unitary

MathlibAnnex.CStarAlgebra.CAR.exists_liftedCornerExponential_apply_eq_involution

theorem

A prescribed unitary involution on a represented root corner can be realized on finitely many vectors by one lifted corner exponential.

Statement

Let A be the completed CAR algebra: the norm completion of the increasing union of $A_n = M_{2^n}(\mathbb{C})$ under the unital embeddings $a \mapsto a \otimes I_2$ (with the fixed coordinate reindexing). Write $\iota_n : A_n \rightarrow A$ for the canonical isometric inclusion and $e_{ij}^{(n)} = \iota_n(E_{ij})$ for its matrix units. Let H be a nonzero complex Hilbert space and $\rho : A \rightarrow B(H)$ an irreducible unital star representation. Fix a stage n , put $e = e_{00}^{(n)}$, and let $R = \text{ran } \rho(e)$. Suppose $U : R \rightarrow R$ is a surjective complex-linear isometry with $U^2 = I_R$. For a finite family $v_0, \dots, v_{d-1} \in R$, there is $h = h^* \in A$, supported by e on both sides, such that the unitary $u = \sum_i e_{i0}^{(n)} (e \exp(ih)) e_{0i}^{(n)}$ satisfies $\rho(u)v_j = Uv_j$ for every j .

Assumptions

The Hilbert space is complete and nonzero; irreducibility here means that the representation has no proper nonzero closed reducing subspace. The integer d may be zero. The selected vectors need not span R . The involution is defined on the whole corner Hilbert space and is unitary there.

Conclusion

The same supported self-adjoint generator gives the ambient unitary and all the prescribed finite actions. Support means $eh = h = he$. The sum amplifies the corner unitary $e \exp(ih)$ to a unitary of the unital algebra A .

Only the displayed finite family is required to have the prescribed action. The theorem does not identify the action on the whole corner with U , and does not give a small-norm bound for $u - 1$. In this Hilbert-operator setting the source adjoint $T^\dagger$ is the ordinary Hilbert adjoint T^* ; the exact source notation is retained.

Proof route

The involution determines an orthogonal projection $Q = (I_R - U)/2$. Let T be the zero extension of πQ from R to H , where π denotes the real circle constant. On the finite span of the selected vectors and their images under U , $\exp(iT)$ acts as U . Supported self-adjoint interpolation realizes this exponential by a corner element; matrix amplification preserves its action on root-corner vectors.

▼ Proof steps

1. Since U is a unitary involution, $Q = (I_R - U)/2$ is an orthogonal projection. Let T be the zero extension of πQ to H . It is self-adjoint, has range in R , and leaves the finite-dimensional space $E = \text{span}\{v_j, Uv_j\}$ invariant. On R , $\exp(i\pi Q) = I_R - 2Q = U$.
2. The supported exponential interpolation result gives a self-adjoint $h \in eAe$ whose corner exponential acts as $\exp(iT)$ on E . Its proof first interpolates T on E with $\|h\| \leq 2\|T\|$, then uses invariance of E to compare the exponentials there.
3. For $z = e \exp(ih)$, the matrix-unit amplification $L_n(z) = \sum_i e_{i0}^{(n)} z e_{0i}^{(n)}$ is unitary. On a root-corner vector its represented action reduces to that of z . Hence $\rho(L_n(z))v_j = Uv_j$. The route explanation below records the accompanying stage-central paths separately.

Main citations

- The stated existence or structural result —
MathlibAnnex.CStarAlgebra.CAR.exists_liftedCornerExponential_apply_eq_involution
- The represented root-corner subspace — MathlibAnnex.CStarAlgebra.CAR.rootCornerSubspace
- Corner exponential realizes the involution on the family —
MathlibAnnex.CStarAlgebra.CAR.exists_rootCornerSupported_exponential_apply_eq_involution
- Ambient amplification preserves root-corner action —
MathlibAnnex.CStarAlgebra.CAR.representation_liftedCornerExponential_apply_of_root
- Supported exact exponential interpolation —
MathlibAnnex.CStarAlgebra.CAR.exists_rootCornerSupported_exponential_eq_on
- The amplified corner exponential — MathlibAnnex.CStarAlgebra.CAR.liftedCornerExponential
- Representation and nonzero irreducibility conventions —
MathlibAnnex.Analysis.CStarAlgebra.Representation.IsIrreducible
- Equivalence of the two irreducibility interfaces on a nonzero Hilbert space —
MathlibAnnex.Analysis.CStarAlgebra.Representation.isIrreducible_iff_starAlgHom

Root-corner route explanation

Lean source signature (exact)

```
theorem exists_liftedCornerExponential_apply_eq_involution
  {H : Type*} [NormedAddCommGroup H] [InnerProductSpace ℂ H]
  [CompleteSpace H] [Nontrivial H]
  (rho : Representation Limit H) (hrho : StarAlgHom.IsIrreducible rho)
  (n d : ℕ) (v : Fin d → rootCornerSubspace rho n)
  (U : rootCornerSubspace rho n ≈i[ℂ] rootCornerSubspace rho n)
  (hU : ∀ x, U (U x) = x) :
  ∃ (h : selfAdjoint Limit)
    (heh : limitMatrixUnit n 0 0 * (h : Limit) = h)
    (hhe : (h : Limit) * limitMatrixUnit n 0 0 = h),
  ∀ i, rho (liftedCornerExponential n h heh hhe : Limit) (v i : H) =
    (U (v i) : H)
```

In the source	Mathematical meaning
H ; <code>[CompleteSpace H]</code> ; <code>[Nontrivial H]</code>	A complete nonzero complex Hilbert space H .
ρ ; <code>hrho : StarAlgHom.IsIrreducible rho</code>	The unital star representation $\rho : A \rightarrow B(H)$ has no proper nonzero closed reducing subspace. Nonzero H is explicit here.
n $d : \mathbb{N}$; <code>rootCornerSubspace rho n</code>	Choose stage n and family length $d \geq 0$. The corner space is $R = \text{ran } \rho(e)$, where $e = e_{00}^{(n)} = \iota_n(E_{00})$.
$v : \text{Fin } d \rightarrow \text{rootCornerSubspace rho n}$	The finite family $v_0, \dots, v_{d-1} \in R$; it need not span R , and $d = 0$ is allowed.
$U : \dots \simeq_{\iota_1} [C] \dots$; <code>hU : $\forall x, U (U x) = x$</code>	A surjective complex-linear isometry $U : R \rightarrow R$ with $U^2 = I_R$, for every $x \in R$.
$\exists (h : \text{selfAdjoint Limit})$	Choose one self-adjoint element $h = h^* \in A$ with both support identities below.
<code>heh : limitMatrixUnit n 0 0 * (h : Limit) = h</code>	The left support identity $eh = h$; <code>(h : Limit)</code> reads the same self-adjoint element as an element of A .
<code>hhe : (h : Limit) * limitMatrixUnit n 0 0 = h</code>	The right support identity $he = h$, for the same h .
<code>liftedCornerExponential n h heh hhe : Limit</code>	The unitary $u = \sum_{k=0}^{2^n-1} e_{k0}^{(n)} (e \exp(ih)) e_{0k}^{(n)} \in A$ built from that h . Here i in $\exp(ih)$ is the imaginary unit.
$\forall i, \rho(\dots) (v\ i : H) = (U (v\ i) : H)$	For every family index $i < d$, $\rho(u)v_i = Uv_i$ in H . The coercions include the same corner vectors in H . All prescribed actions use one h and its unitary u ; no action equality on the whole corner is required.

Further source notes: In the proof, `Complex.I` is the imaginary unit; scalar `pi` and the representation `rho` are different objects.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.CStarAlgebra.CAR.exists_liftedCornerExponential_apply_eq_involution`

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